

Q1

$$\tau = L/R$$

Solution 1: using resistor voltage noise model

Apply same analysis as L9 slide 30 analysis

$$H(s) = Y(s) = 1/Z(s) = 1/(R + sL) = (1/R)/(1 + s\tau)$$

$$I(s) = V_{nR} Y(s)$$

$$I_{no}^2 = \left[4kTR \int_0^\infty |H(f)|^2 df \right]$$

Can calculate using integrals or just use equivalent noise bandwidth of single pole transfer function

$$I_{no}^2 = 4kTR |H(j\omega)|^2 = (4kTR)(1/R)^2 f_{eqL} = (4kTR)(1/R)^2 f_{eqL}$$

$$f_{eqL} = (\pi/2) (1/2\pi)/\tau = 1/(4\tau) = (R/L)/4$$

$$I_{no,rms}^2 = (4kTR)(1/R)^2 (R/L)/4 = kT/L$$

Solution 2: using resistor current noise model

Applying current divider rule

$$H(s) = R/(R + sL) = 1/(1 + s\tau)$$

$$I(s) = I_{nR} H(s) = I_{nR}/(1 + s\tau)$$

$$I_{no}^2 = (4kT/R) |H(j\omega)|^2 = (4kT/R) f_{eqL}$$

$$f_{eqL} = (\pi/2) (1/2\pi)/\tau = (R/L)/4$$

$$I_{no,rms}^2 = (4kT/R)(R/(4L)) = kT/L$$

Using superposition

Case 1 $f_{eqL} < f_{eqC}$

$$V_{on,rms}^2 = I_{no}^2 / G^2 + kT/C = (kT/L)/G^2 + kT/C = (kT/C)\{(C/L)/G^2 + 1\}$$

(4.5 marks)

Case 2 $f_{eqL} > f_{eqC} \Rightarrow$ RC limiting I_{no}^2/G^2

$$\begin{aligned} V_{on,rms}^2 &= I_{no}^2/G^2 (f_{eqC}/f_{eqL}) + kT/C = (kT/L)/G^2 (1/(4RC))/(R/(4L)) + kT/C \\ &= kT/(G^2 R^2 C) + kT/C = (kT/C)\{(1/GR)^2 + 1\} \end{aligned}$$

$$f_{eqL} = (R/L)/4 = R/(4L) = 1/(4\tau)$$

$$f_{eqC} = 1/(4RC) = 1/(4\tau)$$

i) $\tau_C = RC = 10\text{ns}$, $\tau_L = L/R = 1\text{ns} \Rightarrow \omega_C = 1/\tau_C < \omega_L = 1/\tau_L$

$\Rightarrow$ RC limiting. Case 2

$$f_{eqL} = 1/(4n) = 250 \text{ MHz}$$

$$f_{eqC} = 1/(40n) = 1\text{G}/40 = 25 \text{ MHz}$$

$$kT/C = (1.38 \cdot 10^{-23} \cdot 300/10\text{p}) = 0.414 \text{ nV}^2$$

$$V_{on,rms}^2 = (kT/C) \{ (1/GR)^2 + 1 \} = 0.414\text{n} \cdot \{ (1/(1/1\text{k})1\text{k})^2 + 1 \} = 0.414\text{n} \{ 1 + 1 \} = 0.828 \text{ nV}^2$$

$$V_{on,rms} = 28.77 \mu\text{V}$$

(3 marks)

ii) $\tau_C = RC = 1\text{ns}$, $\tau_L = L/R = 10\text{ns} \Rightarrow \omega_C = 1/\tau_C > \omega_L = 1/\tau_L$

$\Rightarrow$ L/R limiting. Case 1

$$f_{eqL} = 1/(40n) = 25 \text{ MHz}$$

$$f_{eqC} = 1/(4n) = 1\text{G}/4 = 250 \text{ MHz}$$

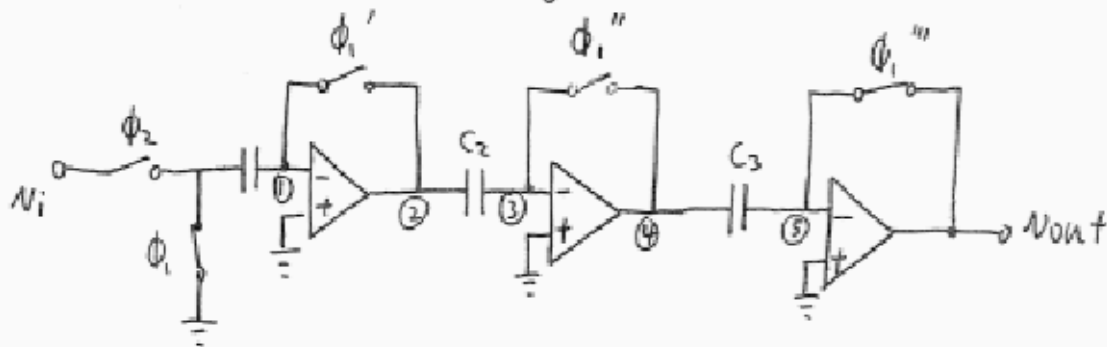
$$kT/C = (1.38 \cdot 10^{-23} \cdot 300/1\text{p}) = 4.14 \text{ nV}^2$$

$$V_{on,rms}^2 = (kT/C) \{ (C/L)/G^2 + 1 \} = (1.38 \cdot 10^{-23} \cdot 300/1\text{p}) \{ (1\text{p}/10\mu)/1\mu + 1 \} = 4.14 \cdot 10^{-9} \{ 0.1 + 1 \} = 4.55 \text{ nV}^2$$

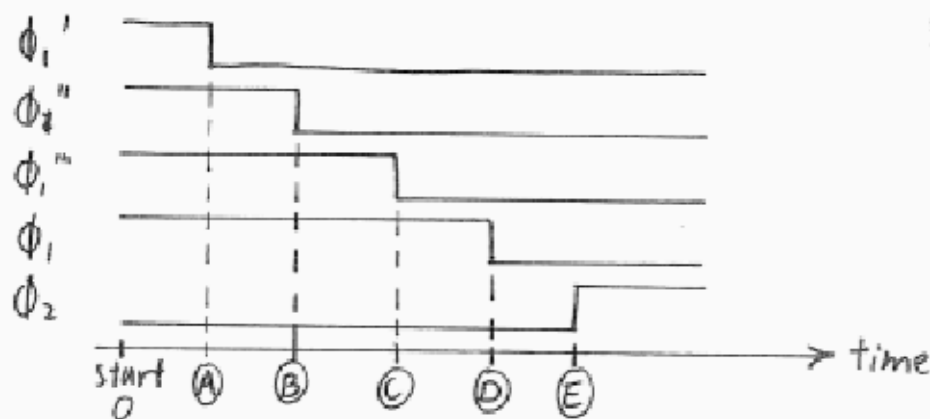
$$V_{on,rms} = 67.45 \mu\text{V}$$

(3 marks)

Show clock feedthrough is $|V_{err-3}|/(A_1 A_2)$



Clocking diagram:



* switches are closed with waveforms high *

Time	$V_{(1)}$	$V_{(2)}$	$V_{(3)}$	$V_{(4)}$	$V_{(5)}$	V_{out}
0	0V	0V	0V	0V	0V	0V
A	V_{err1}	$-A_1 V_{err1}$	0V	0V	0V	0V
B	V_{err1}	$-A_1 V_{err1}$	V_{err2}	$-A_2 V_{err2}$	0V	0V
C	V_{err1}	$-A_1 V_{err1}$	V_{err2}	$-A_2 V_{err2}$	V_{err3}	$-A_3 V_{err3}$
D	"	"	"	"	"	"
E	$V_i + V_{err1}$	$-A_1 (V_i + V_{err1})$	$-A_1 V_i + V_{err2}$	$(A_1 A_2 V_i - A_2 V_{err2})$	$(-A_1 A_2 A_3 V_i + A_3 V_{err3})$	$(A_1 A_2 V_i + V_{err3})$

$\therefore$ at the end of ϕ_2 ,

$$V_{out} = -(A_1 A_2 A_3 V_i + A_3 V_{err3})$$

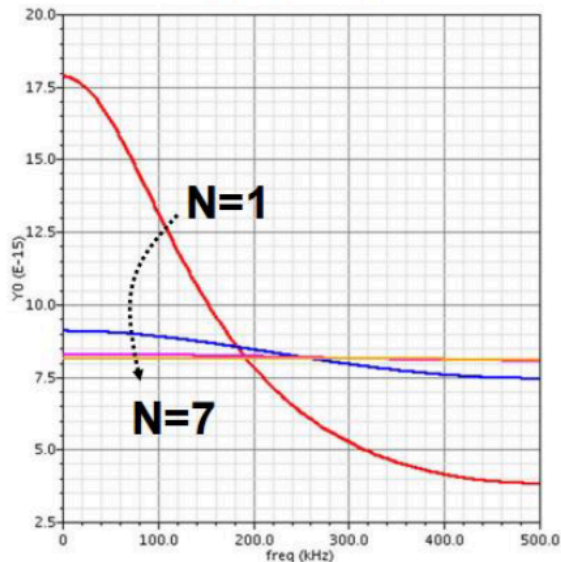
$$= -A_1 A_2 A_3 (V_i + V_{loff}) \text{ where } V_{loff} = \frac{V_{err3}}{A_1 A_2}$$

$\therefore$ the input-referred offset is $\frac{V_{err3}}{A_1 A_2}$. Q.E.D.

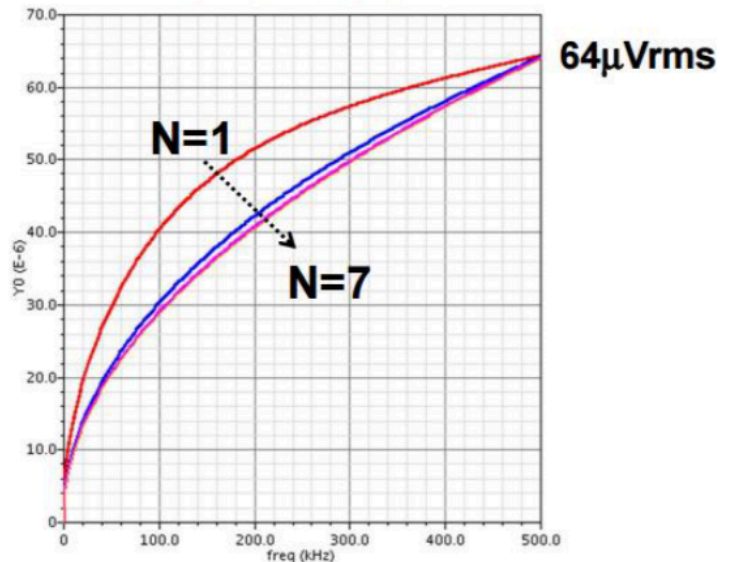
PNOISE Simulation

C=1pF

Noise PSD



Noise Integral



(10 marks) for each type of plot (noise psd, noise integral)

Note that pnoise psd is single-sided spectrum (power only for positive frequencies)

For $N > 5$, noise psd is white which is what we assume for sampling noise psd. (3 marks)

Noise integral $V_{out} = kT/C = 64 \mu V_{rms}$ (2 marks)

- plotting in linear scale allow the shape difference to be more obvious
- the trend of noise integral also
 - they all sum to kT/C , but for those small N , the noise power is bunched up at low frequency, so the noise integral rises faster, for bigger N , its more and more linear